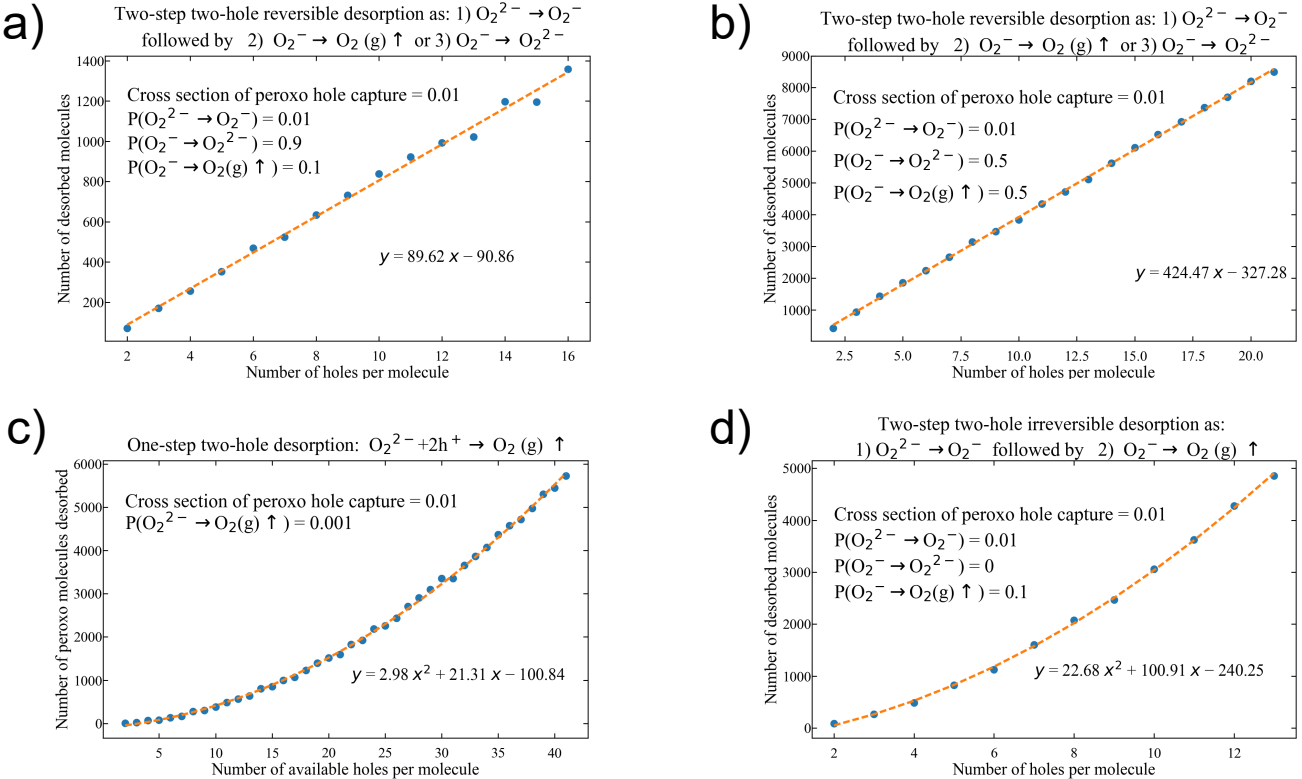

a)
Two-step two-hole reversible desorption as: 1) $O_2^{2-} \rightarrow O_2^-$
followed by 2) $O_2^- \rightarrow O_2$ (g) ↑ or 3) $O_2^- \rightarrow O_2^{2-}$
Cross section of peroxo hole capture = 0.01
$P(O_2^{2-} \rightarrow O_2^-) = 0.01$
$P(O_2^- \rightarrow O_2^{2-}) = 0.9$
$P(O_2^- \rightarrow O_2(g) \uparrow) = 0.1$
$y = 89.62\,x - 90.86$
Number of desorbed molecules
Number of holes per molecule
b)
Two-step two-hole reversible desorption as: 1) $O_2^{2-} \rightarrow O_2^-$
followed by 2) $O_2^- \rightarrow O_2$ (g) ↑ or 3) $O_2^- \rightarrow O_2^{2-}$
Cross section of peroxo hole capture = 0.01
$P(O_2^{2-} \rightarrow O_2^-) = 0.01$
$P(O_2^- \rightarrow O_2^{2-}) = 0.5$
$P(O_2^- \rightarrow O_2(g) \uparrow) = 0.5$
$y = 424.47\,x - 327.28$
Number of desorbed molecules
Number of holes per molecule
c)
One-step two-hole desorption: $O_2^{2-} + 2h^+ \rightarrow O_2$ (g) ↑
Cross section of peroxo hole capture = 0.01
$P(O_2^{2-} \rightarrow O_2(g) \uparrow) = 0.001$
$y = 2.98\,x^2 + 21.31\,x - 100.84$
Number of peroxo molecules desorbed
Number of available holes per molecule
d)
Two-step two-hole irreversible desorption as:
1) $O_2^{2-} \rightarrow O_2^-$ followed by 2) $O_2^- \rightarrow O_2$ (g) ↑
Cross section of peroxo hole capture = 0.01
$P(O_2^{2-} \rightarrow O_2^-) = 0.01$
$P(O_2^- \rightarrow O_2^{2-}) = 0$
$P(O_2^- \rightarrow O_2(g) \uparrow) = 0.1$
$y = 22.68\,x^2 + 100.91\,x - 240.25$
Number of desorbed molecules
Number of holes per molecule